UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	NOVEMBER
TIME	2026

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of 3 questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages

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QUESTION 1

a) Given $f(x) = (5x + 1)/(x - 2)$ and $g(x) = \sqrt{(2x + 6)}$, state the domain of each function.

(4 marks)

b) Given $f(x) = 5 - 3x$ and $g(x) = 5/(x + 5)$, find $(g \circ f)(x)$ and state its domain.

(5 marks)

c) Given $f(x) = (-3x - 2)/(4x + 5)$, $x \neq -5/4$, find $f^{-1}(x)$ and state its domain.

(5 marks)

d) The piecewise function is defined by
$f(x) = 3 - 3x$, if $x < -1$;
$f(x) = -4$, if $-1 \leq x \leq 2$;
$f(x) = x^2 - 3$, if $x > 2$.
Sketch the graph and find $f(-2)$, $f(0)$ and $f(3)$.

(5 marks)

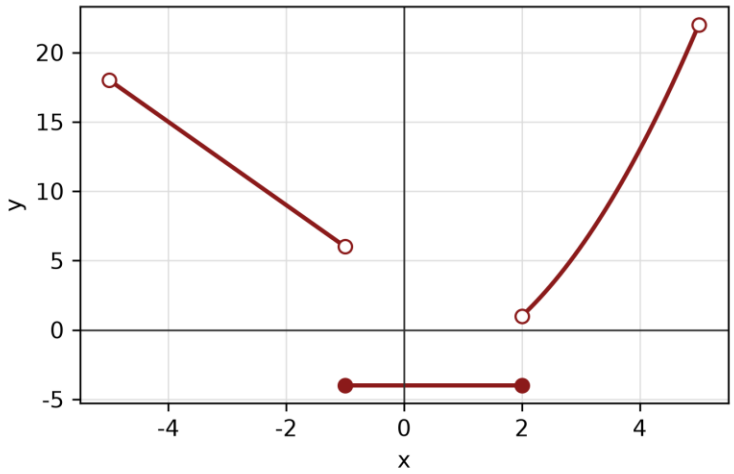


Figure: piecewise graph.

e) Apply graph-transformation techniques to sketch $y = -2(x - 4)^2 - 5$. State the vertex and the y-intercept.

(5 marks)

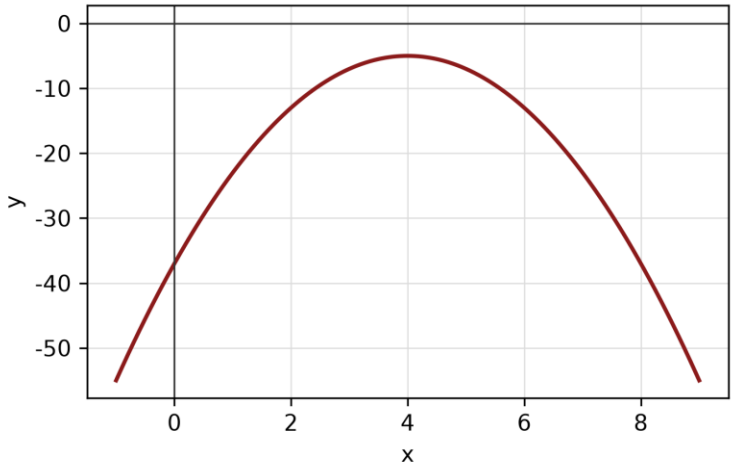


Figure: transformed graph.

f) Determine algebraically whether f(x) = 4x^5 - x^3 + 2x is even, odd or neither. State the symmetry of its graph.

(4 marks)

g) Given f(x) = -5x - 6 and g(x) = 2 - 2x, find (f + g)(x), (f - g)(x) and (f g)(x). State the domain of each.

(4 marks)

h) Sketch y = |x - 2| + 1 for -2 ≤ x ≤ 6. State the domain and range.

(4 marks)

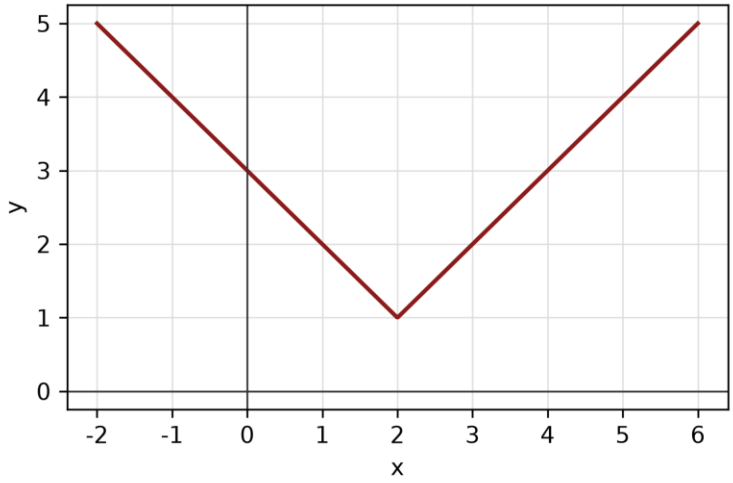


Figure: function graph.

QUESTION 2

a) Solve the inequality -x - 8 < 1. Express the answer in interval notation.

(4 marks)



Figure: solution on a number line.

b) Solve the compound inequality -4 < 5 - 4x < 3. Express the answer in interval notation.

(4 marks)



Figure: solution on a number line.

c) Solve the inequality $-2x^2 + 8x + 10 <= 0$.

(5 marks)

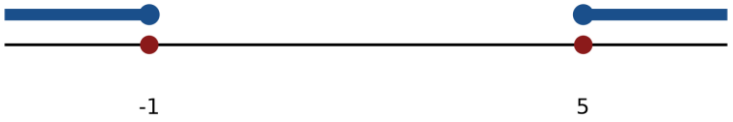


Figure: critical values.

d) Solve the rational inequality (5 - 3x)/(3x - 5) > -1. Express the answer in interval notation.

(5 marks)

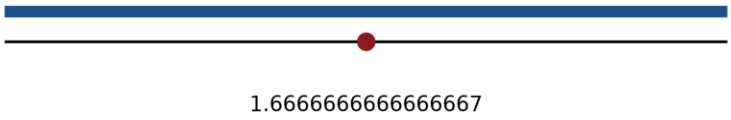


Figure: critical values.

e) Write the solution of 3x - 6 < -2 in interval notation and on a number line.

(3 marks)

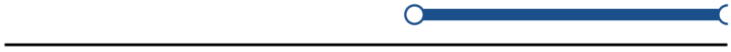


Figure: solution set.

QUESTION 3

a) Solve the equation |-5x - 1| = 5.

(4 marks)

b) Solve the inequality |5 - 4x| >= 7. Express the answer in interval notation and on a number line.

(5 marks)



Figure: solution on a number line.

c) Sketch y = -3|x + 3| + 5. State the vertex, the y-intercept and any x-intercepts.

(5 marks)

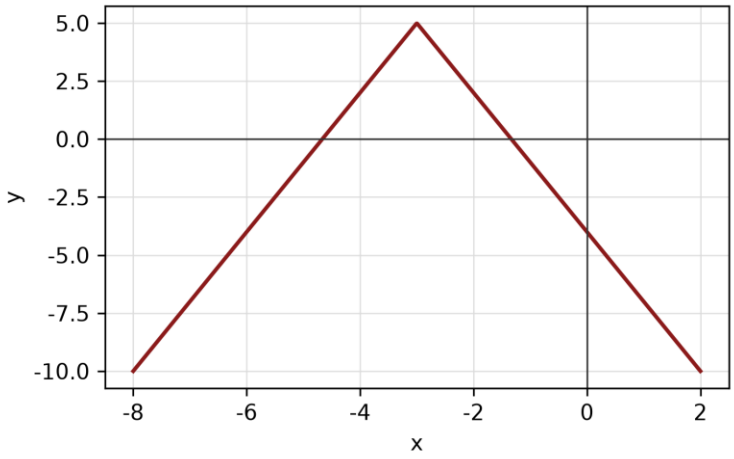


Figure: absolute-value graph.

END OF QUESTION PAPER

APPENDIX 1

Formula sheet (as supplied with the paper).

$\sin^2 A + \cos^2 A = 1$

$\sin(A + B) = \sin A \cos B + \cos A \sin B$

$\cos(A + B) = \cos A \cos B - \sin A \sin B$

$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$

$\cos 2A = \cos^2 A - \sin^2 A$

$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$

$\sin 2A = 2 \sin A \cos A$

$\tan^2 A + 1 = \sec^2 A$

$\tan 2A = 2 \tan A / (1 - \tan^2 A)$

Arc length, $s = r\theta$

Area of sector = $\frac{1}{2}r^2\theta$

$a / \sin A = b / \sin B = c / \sin C$

$a^2 = b^2 + c^2 - 2bc \cos A$

Area of triangle = $\frac{1}{2}ab \sin C$

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